Spectral Reduction in the One-Dimensional Particle-in-a-Box: Verification, Dispersion, and Operator Identifiability

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Abstract

The one-dimensional particle in a box provides a transparent setting for examining what is preserved when a quantum Hamiltonian is discretized and reduced. We apply centered finite differences on closed-interval grids and project the resulting Hamiltonian onto its lowest-energy states, choosing the retained dimension from a prescribed bound on omitted thermal probability. Because both the continuum and discrete spectra are available in closed form, eigensolver accuracy, discretization error, and projection error can be distinguished directly. The computed eigensystem satisfies the finite spectral identities to roundoff. Fixed modes converge to their continuum energies at second order, whereas modes whose wavelengths remain tied to the grid scale retain a finite dispersion error under refinement. Residual comparisons show that consistent compression preserves the zero-potential identity, unmatched compression isolates the discarded kinetic sector, and changing the boundary closure produces a boundary-localized residual rather than a potential. A model-class comparison further shows that a known reduced matrix does not uniquely determine a kinetic–potential decomposition. The familiar box model therefore provides a controlled computational setting for understanding how grid resolution, subspace selection, boundary conditions, and physical assumptions affect the interpretation of reduced Hamiltonians. Separate numerical and machine-checked verification supports these conclusions while remaining distinct from physical validation.

Keywords: particle in a box; spectral reduction; finite differences; operator compression; convergence; identifiability; numerical verification

1 Introduction

The one-dimensional particle in a box is among the first exactly solvable models encountered in quantum mechanics. Its energy levels, standing-wave eigenfunctions, and boundary conditions make quantization visible with little mathematical overhead. It is consequently well established as both a teaching model and a numerical benchmark: software-assisted treatments visualize its states, while finite-difference calculations recover its energies and wavefunctions [13, 9]. That familiarity makes the box useful for a less elementary question: what is actually preserved when a continuum Hamiltonian is represented by a finite matrix and then reduced to a smaller state space?

A numerical calculation of this kind contains several distinct problems. The continuum differential operator has exact energy–wavefunction pairs. A chosen grid and stencil define a different, finite eigenvalue problem. An eigensolver then produces computed approximations to that finite problem. The computed eigenpairs may solve the matrix accurately even when the matrix still approximates

the continuum poorly. Reducing the matrix introduces another choice—the retained subspace—whose accuracy and interpretation must be assessed separately.

Projection-based model reduction provides a general framework for constructing smaller operators on selected trial spaces [2], and Hamiltonian systems motivate reductions that preserve their declared structure [17]. Once a subspace is selected, forming a reduced Hamiltonian is straightforward. Interpreting it is not. A residual matrix can combine discretization, omitted states, a changed boundary realization, or a convention used to divide an operator into kinetic and potential parts. Taking a norm cannot recover meaning that was absent from the subtraction itself.

Constructing and diagnosing reduced operators remains an active area of electronic-structure research. In a common workflow, Kohn–Sham density-functional-theory calculations obtain extended Bloch states in a plane-wave basis, after which maximally localized Wannier-function methods select a band subspace and optimize its gauge [14]. Entangled bands require an additional disentanglement step [23], and WANNIER90 implements these constructions for Bloch states supplied by many electronic-structure codes [15, 18]. Interpreting the resulting localized Hamiltonian is difficult because parent-model approximations, numerical convergence, subspace selection, gauge choice, localization, and real-space truncation can all influence the result. The particle-in-a-box calculation does not model Wannierization; it provides a transparent setting in which the more general meanings of representation, compression, residuals, and identifiability can be established before those effects are combined.

This article is the first stage of a planned diagnostic sequence in which each model introduces one additional structural difficulty. PIAB1D isolates operator compression in an invariant spectral subspace. A one-dimensional quantum harmonic oscillator then introduces spatial-basis truncation. A two-dimensional Bloch model introduces periodicity through twisted boundary conditions, followed by explicit lattice geometry in two and three dimensions, composite and multiband subspaces, and finally a bulk-silicon plane-wave Kohn–Sham calculation reduced to an s-, p-, and d-orbital tight-binding representation. Only the first stage is studied here; the later stages state the intended progression rather than completed evidence. The purpose of the sequence is to preserve diagnostic meaning while increasing physical and numerical complexity one layer at a time.

The box separates these effects unusually cleanly. Both its continuum spectrum and the spectrum of the centered finite-difference Dirichlet matrix are known in closed form, and the lowest represented wavefunctions span an invariant subspace of the finite Hamiltonian. Compression to that subspace is therefore exact for the finite matrix. A discrepancy between the retained discrete energies and their continuum values is finite-difference dispersion, not projection leakage. This controlled case also makes it possible to compare residuals formed with matched projections, unmatched projections, and different boundary closures before moving to systems in which these effects coexist.

The study is organized around four questions:

- How should continuum, represented, and computed energy–wavefunction pairs be distinguished?
- How can a retained dimension M be selected a priori from a declared bound on omitted thermal probability?
- Which diagnostics test finite-dimensional algebra, and which test continuum approximation?
- What can residual matrices and additive kinetic–potential decompositions support physically once their admissible model classes are stated?

Table 1 summarizes the notation used to keep these levels separate.

The intended use is an advanced undergraduate or beginning graduate computational physics case study. The scope remains deliberately narrow: a homogeneous Dirichlet box and its finite

representations. The calculation does not model a finite barrier, scattering, a periodic solid, or a semiconductor, and agreement with the analytical box spectrum is verification against the same mathematical model rather than validation against experiment. A structured prior-art search did not locate one source combining the exact finite spectrum, invariant spectral reduction, continuum-dispersion study, residual semantics, model-class identifiability, separate numerical recomputation, and machine-checked crosswalk used here. The search protocol, source matrix, and limitations are retained in `prior_art_review.md`. The bounded contribution is the integration of these elements into one reproducible calibration, not priority for any individual ingredient.

Object	Continuum	Finite representation or computation
Grid	$x \in [0, L]$	$\boldsymbol{x}_{\Delta x} = [x_1, \ldots, x_N]^{\mathsf{T}}$
Wavefunction	$\psi_n(x)$	$\boldsymbol{\psi}_{n,\Delta x}$, $\widetilde{\boldsymbol{\psi}}_{n,\Delta x}$
Energy	E_n	$E_{n,\Delta x}$, $\widetilde{E}_{n,\Delta x}$
Hamiltonian	H_{box}	$\mathsf{H}_{\Delta x}$, $\mathsf{H}_{\mathrm{red}}$
Retained space	P_M	U_M, P_M, Q_M

Table 1: Continuum, represented, and computed notation. A tilde denotes a computed quantity; the subscript Δx denotes the finite representation. The retained result distinguishes Euclidean-normalized solver vectors from the separately stored quadrature-normalized nodal amplitudes defined in Equation (16).

2 From the continuum box to a finite matrix

We begin with quantum mechanics familiar from undergraduate modern physics and introductory quantum mechanics [1, 8]. The purpose is not to derive a new box solution, but to establish the sequence of representations needed later: an abstract eigenvalue problem, its coordinate-space differential operator, a finite-difference matrix, and finally a reduced operator reconstructed from selected eigenpairs. Keeping these levels explicit is what allows numerical accuracy, continuum accuracy, and reduction accuracy to be discussed separately.

2.1 The time-independent Schrödinger equation

We represent the stationary quantum-mechanical problem by the time-independent Schrödinger equation

$$\widehat{H}|\psi\rangle = E|\psi\rangle. \tag{1}$$

Here $\widehat{H}$ is the Hamiltonian operator, $|\psi\rangle$ is a stationary state, and E is its energy eigenvalue. For one nonrelativistic particle of mass m,

$$\widehat{H} = \frac{\widehat{p}^2}{2m} + \widehat{V}, \qquad \widehat{p} = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}, \tag{2}$$

where $\widehat{p}$ is the momentum operator, $\widehat{V}$ is the potential-energy operator, $\hbar$ is the reduced Planck constant, and $\mathrm{i}^2 = -1$. Since Dirac notation states the eigenvalue problem without choosing a coordinate representation, a concrete representation requires projection onto a basis of interest. Here we choose the position basis. Projecting Equation (1) onto the position bra $\langle x|$ gives

$$\langle x|\widehat{H}|\psi\rangle = \langle x|E|\psi\rangle = E\langle x|\psi\rangle. \tag{3}$$

With $\psi(x) = \langle x|\psi\rangle$, we obtain the familiar coordinate-space Schrödinger equation

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right]\psi(x) = E\psi(x). \tag{4}$$

This is a partial differential equation in its general spatial form and an ordinary differential equation in the present one-dimensional setting.

2.2 The continuum particle in a box

For the particle in a box, $V(x) = 0$ on the open interval $(0, L)$ and the wavefunction satisfies

$$\psi(0) = \psi(L) = 0. \tag{5}$$

The boundary conditions quantize the solutions. Labeling them by the positive integer n, the normalized eigenfunctions and energies are

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\!\left(\frac{n\pi x}{L}\right), \qquad E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}, \qquad n \geq 1. \tag{6}$$

This is the closed-form spectrum of the standard Dirichlet box problem [26]. Figure 1a shows the first three continuum eigenfunctions as an energy ladder. Each wavefunction is displaced vertically so that its zero-amplitude line lies at the corresponding energy E_n. The displacement uses one common visual scale and does not give the wavefunction units of energy.

2.3 Discrete representation

We now discretize the coordinate-space Schrödinger equation in Equation (4), which follows from the position projection in Equation (3). First decompose the Hamiltonian into its kinetic- and potential-energy operators,

$$\widehat{H} = \widehat{T} + \widehat{V}, \qquad \widehat{T} = \frac{\widehat{p}^2}{2m}. \tag{7}$$

The projected eigenvalue problem can then be written as

$$\begin{aligned}\langle x|\widehat{H}|\psi\rangle &= \langle x|(\widehat{T} + \widehat{V})|\psi\rangle &(8)\\ &= \langle x|\widehat{T}|\psi\rangle + \langle x|\widehat{V}|\psi\rangle &(9)\\ &= E\langle x|\psi\rangle. &(10)\end{aligned}$$

In the position representation,

$$\langle x|\widehat{T}|\psi\rangle = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}, \qquad \langle x|\widehat{V}|\psi\rangle = V(x)\psi(x). \tag{11}$$

The potential term acts by multiplication, while the kinetic term contains the second derivative that must be represented on the grid. For the particle in a box, $V(x) = 0$ at every interior point.

Discretize the closed interval $x \in [0, L]$ with N ordered points. Define

$$\boldsymbol{x}_{\Delta x} = [x_1, x_2, \ldots, x_N]^{\mathsf{T}}, \qquad x_i = (i-1)\Delta x, \qquad \Delta x = \frac{L}{N-1},$$

and sample the wavefunction as

$$\boldsymbol{\psi}_{\Delta x} = [\psi_1, \psi_2, \ldots, \psi_N]^{\mathsf{T}}, \qquad \psi_i = \psi(x_i).$$

The Dirichlet boundary conditions set $\psi_1 = \psi_N = 0$. The kinetic matrix therefore acts on the interior restriction

$$\boldsymbol{x}^{\circ}_{\Delta x} = [x_2, \ldots, x_{N-1}]^{\mathsf{T}}, \qquad \boldsymbol{\psi}^{\circ}_{\Delta x} = [\psi_2, \ldots, \psi_{N-1}]^{\mathsf{T}} \in \mathbb{R}^{N-2}.$$

Taylor expansion about an interior point x_i, with $i = 2, \ldots, N-1$, gives

$$\left.\frac{\mathrm{d}\psi}{\mathrm{d}x}\right|_{x_i} = \frac{\psi_{i+1} - \psi_{i-1}}{2\Delta x} + \mathcal{O}\big((\Delta x)^2\big)\,,$$
$$\left.\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\right|_{x_i} = \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{(\Delta x)^2} + \mathcal{O}\big((\Delta x)^2\big)\,.$$

The first formula represents the derivative appearing in the momentum operator. The second directly represents the kinetic-energy operator. Together with pointwise sampling of the potential on the interior grid, they define the discrete map

$$\widehat{H} = \widehat{T} + \widehat{V} \quad \longmapsto \quad \mathsf{H}_{\Delta x} = \mathsf{T}_{\Delta x} + \mathsf{V}_{\Delta x}, \qquad \mathsf{V}_{\Delta x} = \mathrm{diag}\big(V(x_2), \ldots, V(x_{N-1})\big).$$

The matrices $\mathsf{T}_{\Delta x}$, $\mathsf{V}_{\Delta x}$, and $\mathsf{H}_{\Delta x}$ belong to $\mathbb{R}^{(N-2)\times(N-2)}$ and act on $\boldsymbol{\psi}^{\circ}_{\Delta x}$. This line marks the change in notation. Before discretization, hatted symbols denote continuum operators. From this point onward, upright sans-serif capitals such as H and R denote finite matrices, bold symbols such as $\boldsymbol{\psi}$ denote finite vectors, and italic symbols such as E and δ denote scalars. Abstract operators that reappear in the spectral discussion remain italic and are identified by their domains or Dirac notation. Thus R denotes a residual matrix, while $\delta = \|\mathsf{R}\|_{\mathrm{F}}$ denotes a scalar diagnostic derived from it. Since $V(x_i) = 0$ at every interior point of the box, $\mathsf{V}_{\Delta x} = \mathsf{0}$.

Equivalently, the continuum box Hamiltonian is the operator

$$\widehat{H}_{\mathrm{box}} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}, \qquad \mathcal{D}(\widehat{H}_{\mathrm{box}}) = H^2(0, L) \cap H^1_0(0, L). \tag{12}$$

The boundary condition belongs to the operator domain; it is not a finite-valued potential at the endpoints. Applying the centered second-derivative stencil on the $N-2$ interior points gives

$$\mathsf{H}_{\Delta x} = \frac{\hbar^2}{2m(\Delta x)^2}\begin{bmatrix} 2 & -1 & & 0 \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ 0 & & -1 & 2 \end{bmatrix} \in \mathbb{R}^{(N-2)\times(N-2)}. \tag{13}$$

Let $\boldsymbol{\psi}_{n,\Delta x} \in \mathbb{R}^{N-2}$ denote the nth normalized represented wavefunction. The finite matrix eigenvalue problem is

$$\mathsf{H}_{\Delta x}\boldsymbol{\psi}_{n,\Delta x} = E_{n,\Delta x}\boldsymbol{\psi}_{n,\Delta x}, \qquad n = 1, \ldots, N-2, \tag{14}$$

where the represented wavefunctions are discrete sine modes and

$$E_{n,\Delta x} = \frac{2\hbar^2}{m(\Delta x)^2}\sin^2\!\left(\frac{n\pi}{2(N-1)}\right). \tag{15}$$

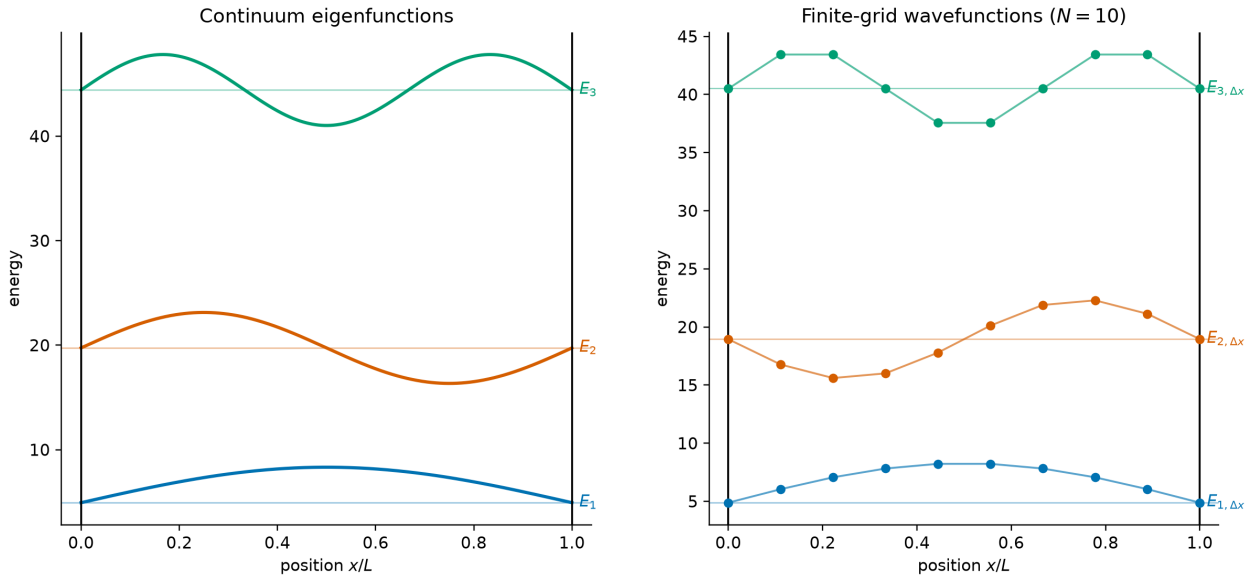
The eigensolver stores each represented wavefunction in Euclidean normalization, $\boldsymbol{\psi}^{\mathsf{T}}_{n,\Delta x}\boldsymbol{\psi}_{n,\Delta x} = 1$. For the uniform interior grid, the quadrature matrix is the constant diagonal matrix $\mathsf{W}_{\Delta x} = \Delta x\,\mathsf{I}_{N-2}$. The corresponding nodal amplitudes in quadrature normalization are therefore

$$\boldsymbol{\psi}^{(q)}_{n,\Delta x} = (\Delta x)^{-1/2}\boldsymbol{\psi}_{n,\Delta x}, \qquad \big(\boldsymbol{\psi}^{(q)}_{n,\Delta x}\big)^{\mathsf{T}}\mathsf{W}_{\Delta x}\boldsymbol{\psi}^{(q)}_{n,\Delta x} = 1. \tag{16}$$

For the canonical sine convention, this rescaling gives $\psi^{(q)}_{n,\Delta x,j} = \sqrt{2/L}\sin(n\pi j/(N-1)) = \psi_n(x_j)$ at every interior node. If $\mathsf{U}^{(q)}_M = (\Delta x)^{-1/2}\mathsf{U}_M$, then

$$\left(\mathsf{U}^{(q)}_M\right)^{\mathsf{T}}\mathsf{W}_{\Delta x}\mathsf{U}^{(q)}_M = \mathsf{I}_M, \quad \mathsf{U}^{(q)}_M\left(\mathsf{U}^{(q)}_M\right)^{\mathsf{T}}\mathsf{W}_{\Delta x} = \mathsf{U}_M\mathsf{U}_M^{\mathsf{T}}, \quad \left(\mathsf{U}^{(q)}_M\right)^{\mathsf{T}}\mathsf{W}_{\Delta x}\mathsf{H}_{\Delta x}\mathsf{U}^{(q)}_M = \mathsf{U}_M^{\mathsf{T}}\mathsf{H}_{\Delta x}\mathsf{U}_M. \tag{17}$$

Thus the constant quadrature weight changes the displayed nodal amplitudes but not the projector or reduced coordinate matrix used below. The finite-grid wavefunctions in Figure 1b use the quadrature-normalized amplitudes from Equation (16). The points at $x = 0$ and $x = L$ display the homogeneous Dirichlet boundary values and are not interior degrees of freedom.



(a) Continuum eigenfunctions centered on E_n.

(b) Finite-grid wavefunctions centered on $E_{n,\Delta x}$.

Figure 1: The first three PIAB1D states shown as energy ladders. The vertical wavefunction displacement is scaled for visibility; the horizontal reference line in each trace marks its associated energy.

Equation (13) is the usual centered approximation to Equation (12) [24]. The spectra in Equations (6) and (15) are exact solutions of different eigenvalue problems.

2.4 Spectral reconstruction

On the mathematical side, the spectral theorem reconstructs a self-adjoint operator H from its projection-valued measure P_H [19, 26]:

$$I = \int_{\mathbb{R}} \mathrm{d}P_H(\lambda), \qquad H = \int_{\mathbb{R}} \lambda\,\mathrm{d}P_H(\lambda). \tag{18}$$

The second integral is interpreted on the operator domain of H. In the physics formulation, an observable A with a pure-point spectrum has the spectral reconstruction

$$I = \sum_a P_a, \qquad A = \sum_a a P_a, \tag{19}$$

where P_a projects onto the eigenspace with measured value a. For a nondegenerate spectrum, $P_a = |a\rangle\langle a|$. This is the standard operator construction from eigenvalues and eigenstates used by Shankar and by Sakurai and Napolitano [21, 20]. Specializing it to the box gives

$$I = \sum_{n=1}^{\infty} |\psi_n\rangle\langle\psi_n|, \qquad H_{\text{box}} = \sum_{n=1}^{\infty} E_n |\psi_n\rangle\langle\psi_n|. \tag{20}$$

The second equality is understood through the action of the spectral series on states in $\mathcal{D}(H_{\text{box}})$. The finite matrix has the corresponding complete spectral reconstruction [10, 7]:

$$\mathsf{H}_{\Delta x} = \sum_{n=1}^{N-2} E_{n,\Delta x} \boldsymbol{\psi}_{n,\Delta x} \boldsymbol{\psi}_{n,\Delta x}^{\mathsf{T}}. \tag{21}$$

2.5 Operator reduction

The time-independent Schrödinger equation identifies the energy eigenstates and their energies:

$$H_{\text{box}} |\psi_n\rangle = E_n |\psi_n\rangle. \tag{22}$$

A state in the operator domain can be expanded in the complete energy basis as

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |\psi_n\rangle, \qquad c_n = \langle\psi_n|\psi\rangle. \tag{23}$$

Applying the Hamiltonian gives

$$H_{\text{box}} |\psi\rangle = \sum_{n=1}^{\infty} E_n c_n |\psi_n\rangle. \tag{24}$$

This componentwise action is the physics form of the spectral reconstruction [21, 20]. Thus E_n is the spectral multiplier that scales the nth component under the action of the Hamiltonian. The coefficient c_n is the state's expansion amplitude, and $|c_n|^2$ is its measurement probability when the state is normalized; the magnitude of E_n is not itself a probability weight. Energy eigenvalues enter probability assignments only after a statistical ensemble has been specified. In the canonical ensemble, the equilibrium probability of the nondegenerate energy state n is

$$p_n = \frac{\exp(-\beta E_n)}{Z}, \qquad Z = \sum_j \exp(-\beta E_j), \qquad \beta = \frac{1}{k_{\text{B}} T}, \tag{25}$$

as described by Pathria and Beale [16]. These Boltzmann probabilities depend on the chosen ensemble and temperature and are distinct from the Born probabilities $|c_n|^2$ of a specified pure state.

We use this interpretation to motivate basis truncation in the Hamiltonian reduction. For a pure state, the probability omitted above mode M is $\sum_{n>M} |c_n|^2$. For a canonical ensemble, the corresponding omitted thermal probability is

$$w_{>M}^{(\beta)} = \frac{1}{Z} \sum_{n>M} \exp(-\beta E_n). \tag{26}$$

A low-energy truncation is therefore appropriate when the intended states have little amplitude outside the retained subspace or when the omitted thermal weight is small at the temperature of interest.

A reduction selects a spectral subset and restricts the operator to its span. For the lowest M box states, define

$$P_M = \sum_{n=1}^{M} |\psi_n\rangle\langle\psi_n|, \qquad H^{(M)} = P_M H_{\text{box}} P_M = \sum_{n=1}^{M} E_n |\psi_n\rangle\langle\psi_n|. \tag{27}$$

The finite calculation applies the same construction to the complete matrix spectrum. Let $\mathsf{U}_M = [\boldsymbol{\psi}_{1,\Delta x} \ \cdots \ \boldsymbol{\psi}_{M,\Delta x}] \in \mathbb{R}^{(N-2)\times M}$ contain the lowest M normalized represented wavefunctions of $\mathsf{H}_{\Delta x}$. The spectral projector, its complement, and the reduced Hamiltonian are

$$\mathsf{P}_M = \mathsf{U}_M \mathsf{U}_M^{\mathsf{T}}, \qquad \mathsf{Q}_M = \mathsf{I}_{N-2} - \mathsf{P}_M, \qquad \mathsf{H}_{\text{red}} = \mathsf{U}_M^{\mathsf{T}} \mathsf{H}_{\Delta x} \mathsf{U}_M. \tag{28}$$

Define the diagonal matrix

$$\mathsf{D}_{\Delta x} = \operatorname{diag}(E_{1,\Delta x}, \ldots, E_{M,\Delta x}). \tag{29}$$

Because the columns of U_M are represented eigenfunctions,

$$\mathsf{H}_{\Delta x}\mathsf{U}_M = \mathsf{U}_M \mathsf{D}_{\Delta x}, \qquad \mathsf{H}_{\text{red}} = \mathsf{D}_{\Delta x}, \qquad \mathsf{Q}_M \mathsf{H}_{\Delta x} \mathsf{U}_M = \mathsf{0}. \tag{30}$$

These are finite-dimensional spectral identities [10, 7]. The last equality states that the selected finite eigenspace has no leakage into its orthogonal complement.

The embedded operator $\mathsf{P}_M \mathsf{H}_{\Delta x} \mathsf{P}_M$ is an $(N-2) \times (N-2)$ matrix. The coordinate matrix H_{red} is an $M \times M$ matrix. They describe the same restriction in different representations, but they cannot be subtracted without an explicit map between those representations.

A similar distinction separates the discrete and continuum spectral references. Set

$$\mathsf{D} = \operatorname{diag}(E_1, \ldots, E_M). \tag{31}$$

Then

$$\mathsf{H}_{\Delta x}\mathsf{U}_M - \mathsf{U}_M \mathsf{D}_{\Delta x} = \mathsf{0}, \tag{32}$$

whereas

$$\mathsf{H}_{\Delta x}\mathsf{U}_M - \mathsf{U}_M \mathsf{D} = \mathsf{U}_M(\mathsf{D}_{\Delta x} - \mathsf{D}). \tag{33}$$

The second residual measures finite-difference dispersion. It is not a failure of the projection.

The retained dimension M is therefore an explicit resolution parameter of the operator reduction. We now describe a diagnostic methodology for choosing M. The method compares candidate retained subspaces through omitted spectral weight, retained-mode dispersion, spectral leakage, and operator-residual norms, then applies a declared selection rule to the admissible candidates.

3 Methodology

Because PIAB1D has both an analytical continuum solution and an exact discrete representation, it provides a clean sandbox for studying operator reduction by spectral reduction. The calculation separates diagnostics of the eigensolver, the finite-difference approximation, the spectral projection, and the interpretation of operator residuals.

3.1 Calculation design

All calculations use the nondimensional convention $L = m = \hbar = 1$. The core calculation uses $N = 10$ points on the closed interval, hence eight interior degrees of freedom. The canonical-weight calculation in Section 4.1 selects the lowest $M = 3$ represented energy–wavefunction pairs. The grid studies evaluate every closed-interval grid size

$$N = 10, 11, \ldots, 258, \tag{34}$$

which corresponds to every interior dimension $N - 2 = 8, 9, \ldots, 256$. The eigensystems are computed in binary64 arithmetic by the pinned PhysKit PIAB1D finite-difference solver. Quadrature handling is enabled explicitly in the retained input. The result stores the Euclidean-normalized solver vectors separately from the quadrature-normalized nodal amplitudes, together with the endpoint-inclusive trapezoidal-rule identity, spacing, full and interior weights, and diagonal mass matrix. The unit-free rule and its homogeneous-Dirichlet restriction are supplied by Project Koios Numerics, while PhysKit applies the weights to the sampled quantum states. Because the present interior quadrature matrix is a scalar multiple of the identity, Equation (17) shows that this rescaling leaves every projector, reduced matrix, and residual reported here unchanged. The Hamiltonian is still solved as a standard real-symmetric eigenproblem, not as a generalized-mass-matrix eigenproblem. A nonuniform grid, nonconstant rule, or consistent finite-element mass matrix is outside the present calculation.

Separate verification scripts recompute the closed discrete spectrum, spectral reconstruction, projector identities, retained embeddings, and residual formulas.

3.2 Implemented reduction representations

For the core calculation, define $\mathbf{\Phi}_M := \mathsf{U}_M \in \mathbb{R}^{8\times 3}$. This reduction basis is distinct from the complementary projector $\mathsf{Q}_M = \mathsf{I}_{N-2} - \mathsf{P}_M$ in Equation (28). The implementation binds the aligned representations in one `ReducedHamiltonian` object:

`spectral_subspace` An `OrthogonalSpectralSubspace` whose `basis_vectors` are $\mathbf{\Phi}_M$, an 8×3 matrix whose columns are the three selected eigenvectors. Its `projector()` method derives $\mathsf{P}_M = \mathbf{\Phi}_M \mathbf{\Phi}_M^{\mathsf{T}}$, the 8×8 orthogonal projector onto the selected subspace.

`matrix` $\mathbf{\Phi}_M^{\mathsf{T}} \mathsf{H}_{\Delta x} \mathbf{\Phi}_M$, the 3×3 reduced Hamiltonian used by the decomposition.

`embedded.matrix` $\mathbf{\Phi}_M (\mathbf{\Phi}_M^{\mathsf{T}} \mathsf{H}_{\Delta x} \mathbf{\Phi}_M) \mathbf{\Phi}_M^{\mathsf{T}}$, the 8×8 reduced Hamiltonian mapped back onto the full grid space.

The abstract `SpectralSubspace` contract does not assert that every Hamiltonian has an orthogonal spectral projector. PIAB1D uses the concrete orthogonal case because its represented Hamiltonian is real symmetric. The reduced and embedded matrices represent the same restricted operator on different coordinate spaces. The projector records which full-space states belong to that reduction; the complementary projector remains $\mathsf{Q}_M = \mathsf{I}_{N-2} - \mathsf{P}_M$.

Three numerical questions are considered. The first is whether the computed finite matrix agrees with the exact discrete formulas. The second is how fixed and growing mode indices behave as the grid is refined. The third is how the meaning of a residual changes when compression, boundary closure, norm, or admissible model class changes.

3.3 Spectral and reduction diagnostics

3.3.1 Computed energy–wavefunction inventory

Let $(\widetilde{E}_{n,\Delta x}, \widetilde{\boldsymbol{\psi}}_{n,\Delta x})$ denote the computed represented eigenpair, ordered by increasing energy. Collect the complete computed wavefunction basis and its energies as

$$\mathsf{U} = [\widetilde{\boldsymbol{\psi}}_{1,\Delta x} \ \cdots \ \widetilde{\boldsymbol{\psi}}_{N-2,\Delta x}], \qquad \widetilde{\mathsf{D}}_{\Delta x} = \operatorname{diag}(\widetilde{E}_{1,\Delta x}, \ldots, \widetilde{E}_{N-2,\Delta x}). \tag{35}$$

Thus $\mathsf{U} \in \mathbb{R}^{(N-2)\times(N-2)}$ contains every computed represented wavefunction, while $\widetilde{\mathsf{D}}_{\Delta x}$ contains the corresponding computed energies. All matrix residuals below use the Frobenius norm

$$\|\mathsf{A}\|_{\mathrm{F}} = \left(\sum_{i,j} |A_{ij}|^2 \right)^{1/2}. \tag{36}$$

It aggregates every matrix-entry discrepancy into one nonnegative scalar.

3.3.2 Closed-form discrete-spectrum agreement

The discrete-spectrum diagnostic is

$$\delta_E = \max_{1 \le n \le N-2} \left| \widetilde{E}_{n,\Delta x} - E_{n,\Delta x} \right|. \tag{37}$$

For each mode, the calculation subtracts the exact represented energy in Equation (15) from the corresponding computed energy, takes the absolute value, and reports the largest discrepancy. This tests whether the eigensolver and the assembled finite matrix reproduce the known spectrum of the discrete problem. It does not compare the discrete spectrum with the continuum energies E_n.

3.3.3 Full spectral reconstruction

The complete computed eigensystem is used to reconstruct the represented Hamiltonian, and the discrepancy is measured by

$$\delta_{\mathrm{rec}} = \left\| \mathsf{H}_{\Delta x} - \mathsf{U}\widetilde{\mathsf{D}}_{\Delta x}\mathsf{U}^{\mathsf{T}} \right\|_{\mathrm{F}}. \tag{38}$$

The calculation forms the matrix product $\mathsf{U}\widetilde{\mathsf{D}}_{\Delta x}\mathsf{U}^{\mathsf{T}}$, subtracts it from the assembled Hamiltonian, and evaluates the Frobenius norm of the difference. A small δ_{rec} checks the completeness and mutual consistency of the computed energies and represented wavefunctions. It also detects mismatched energy–wavefunction ordering. It does not establish that $\mathsf{H}_{\Delta x}$ accurately represents the continuum operator.

3.3.4 Projector idempotency

The retained represented wavefunctions define $\mathsf{P}_M = \mathsf{U}_M\mathsf{U}_M^{\mathsf{T}}$. Its idempotency defect is

$$\delta_{\mathrm{idem}} = \|\mathsf{P}_M^2 - \mathsf{P}_M\|_{\mathrm{F}}. \tag{39}$$

The calculation applies the projector twice, subtracts one application, and measures the remaining matrix. An exact projector satisfies $\mathsf{P}_M^2 = \mathsf{P}_M$, so this diagnostic checks that the computed retained basis produces a numerically valid projection.

3.3.5 Retained coordinate representation

The reduced coordinate matrix should equal the diagonal matrix of the selected exact discrete eigenvalues. The diagnostic is

$$\delta_{\mathrm{coord}} = \|\mathsf{H}_{\mathrm{red}} - \mathsf{D}_{\Delta x}\|_{\mathrm{F}}. \tag{40}$$

Here the calculation constructs $\mathsf{H}_{\mathrm{red}} = \mathsf{U}_M^{\mathsf{T}}\mathsf{H}_{\Delta x}\mathsf{U}_M$, subtracts $\mathsf{D}_{\Delta x} = \mathrm{diag}(E_{1,\Delta x}, \ldots, E_{M,\Delta x})$, and measures the difference. A small value confirms that the retained basis diagonalizes the represented Hamiltonian with the expected discrete energies.

3.3.6 Retained embedding

The same reduced operator can be represented either directly in the full coordinate space or lifted from retained coordinates. Their discrepancy is

$$\delta_{\mathrm{embed}} = \left\| \mathsf{P}_M\mathsf{H}_{\Delta x}\mathsf{P}_M - \mathsf{U}_M\mathsf{H}_{\mathrm{red}}\mathsf{U}_M^{\mathsf{T}} \right\|_{\mathrm{F}}. \tag{41}$$

The first term compresses the Hamiltonian in the full $(N-2)$-dimensional coordinate space. The second term maps the $M \times M$ reduced coordinate matrix back into that space. This diagnostic is important because matrices of different dimensions cannot be compared until an explicit embedding has placed them in the same representation.

3.3.7 Spectral leakage

The leakage diagnostic is

$$\delta_{\mathrm{leak}} = \|\mathsf{Q}_M\mathsf{H}_{\Delta x}\mathsf{U}_M\|_{\mathrm{F}}. \tag{42}$$

The product $\mathsf{H}_{\Delta x}\mathsf{U}_M$ applies the Hamiltonian to every retained basis vector, and left multiplication by Q_M extracts the component outside the retained subspace. A zero value means that the subspace is invariant under the represented Hamiltonian. This vanishes analytically for a subspace spanned by exact represented wavefunctions; it need not vanish for a general trial subspace.

Together these diagnostics test distinct finite-dimensional identities. None of them measures error with respect to the continuum operator; that comparison requires the dispersion diagnostics introduced next.

3.4 Continuum-dispersion diagnostics

The relative energy error of mode n is

$$\epsilon_{n,N} = \frac{\left|\widetilde{E}_{n,\Delta x} - E_n\right|}{E_n}. \tag{43}$$

For two successive grids with spacings Δx_k and Δx_{k+1}, the observed order is

$$p_{n,k} = \frac{\log(\epsilon_{n,k}/\epsilon_{n,k+1})}{\log(\Delta x_k/\Delta x_{k+1})}. \tag{44}$$

Fixed-n sequences test pointwise spectral convergence. Sequences with $q = n/(N-1)$ approximately fixed test modes whose index grows with the grid. These are different convergence questions. Such high-frequency nonuniformity is a known feature of finite-difference wave models [29]; the exact PIAB1D spectrum lets us exhibit it directly for a stationary eigenvalue problem.

3.5 Operator-residual and model-class diagnostics

Residual-based a posteriori estimators can relate a reduced-model residual to an error when the governing problem supplies the necessary stability assumptions and constants [6]. The residuals here have a different purpose: they distinguish which finite operators have been subtracted and are not asserted to be general error bounds.

Three residual matrices are evaluated. Consistent compression compares the Hamiltonian and kinetic matrix after applying the same projector,

$$\mathsf{R}_{\mathrm{cons}} = \mathsf{P}_M \mathsf{H}_{\Delta x} \mathsf{P}_M - \mathsf{P}_M \mathsf{T}_{\mathrm{D},\Delta x} \mathsf{P}_M. \tag{45}$$

Unmatched compression compares the embedded retained Hamiltonian with the uncompressed kinetic matrix,

$$\mathsf{R}_{\mathrm{un}} = \mathsf{P}_M \mathsf{H}_{\Delta x} \mathsf{P}_M - \mathsf{T}_{\mathrm{D},\Delta x}, \tag{46}$$

and the boundary-closure residual is

$$\mathsf{R}_{\mathrm{bc}} = \mathsf{H}_{\mathrm{D},\Delta x} - \mathsf{H}_{\mathrm{cyclic},\Delta x}. \tag{47}$$

Each residual is measured with the Frobenius, spectral, and maximum-entry norms. The same-grid normalized diagnostic is

$$\eta_*(\mathsf{R}) = \frac{\|\mathsf{R}\|_*}{\|\mathsf{H}_{\Delta x}\|_*}, \qquad * \in \{\mathrm{F}, 2, \max\}. \tag{48}$$

These norms describe different matrix properties and are not interchangeable.

Hamiltonian identification usually asks whether unknown dynamical parameters can be recovered from declared controls and measurements [12, 28]. Operator inference instead learns reduced operators from dynamical data, including under Hamiltonian structure [22, 11]. The present question is narrower and static: once a reduced real symmetric matrix is known, does that matrix alone determine a unique additive decomposition within a declared pair of component classes? To test this distinction, the fit of an illustrative shift K to an admissible potential-matrix class $\mathcal{C}$ is summarized by

$$\rho_{\mathcal{C}}(\mathsf{K}) = \min_{\mathsf{V} \in \mathcal{C}} \|\mathsf{K} - \mathsf{V}\|_{\mathrm{F}}. \tag{49}$$

A zero fit residual establishes membership in the declared matrix class; it does not establish a physical interpretation.

3.6 Diagnostic selection of the retained dimension

The retained dimension M is selected by requiring the reduced basis to retain a prescribed fraction of a declared canonical ensemble. On the finite N-point representation, define

$$Z_{\Delta x} = \sum_{j=1}^{N-2} \exp(-\beta E_{j,\Delta x}), \qquad w_{>M}^{(\beta,\Delta x)} = \frac{1}{Z_{\Delta x}} \sum_{n=M+1}^{N-2} \exp(-\beta E_{n,\Delta x}). \tag{50}$$

The omitted-probability tolerance τ is declared a priori, before any candidate retained dimension is evaluated. The required retained dimension is then

$$M_\star(N, \beta, \tau) = \min \left\{ M \in \{1, \ldots, N-2\} : w_{>M}^{(\beta,\Delta x)} \le \tau \right\}. \tag{51}$$

The worked calculation declares the core grid $N = 10$, inverse temperature $\beta = 0.1$ in inverse nondimensional energy units, and tolerance $\tau = 0.01$ before computing the candidate weights. The calculation then returns the smallest low-energy basis that retains at least 99% of the declared canonical probability. The remaining diagnostics test the numerical and continuum quality of the selected reduction.

4 Results and discussion

4.1 An a priori tolerance selects a stable dimension

Figure 2 turns the selection rule into a decision map. Because the retained dimension is discrete, $M_\star(\tau)$ consists of horizontal plateaus separated by the canonical tail weights. The declared tolerance $\tau = 0.01$ lies inside the $M_\star = 3$ plateau: one and two modes are insufficient, whereas the third mode is the first to satisfy the requirement. Thus $M = 3$ is an output of the declared ensemble and tolerance, not a dimension chosen after inspecting the subsequent residuals.

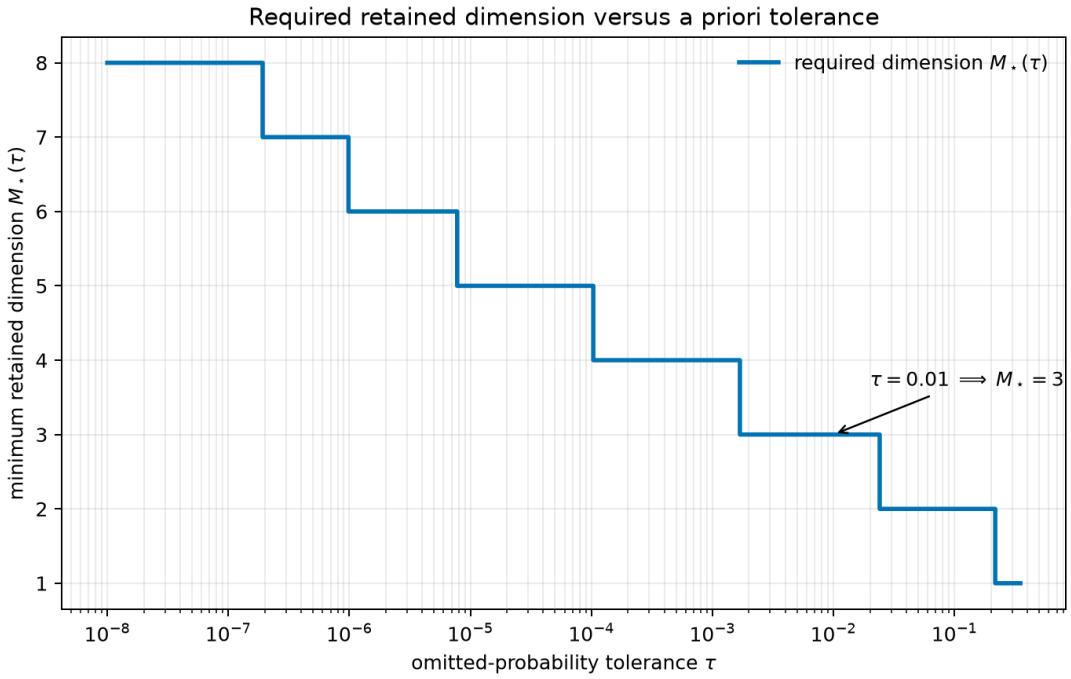


Figure 2: Minimum retained dimension as a function of the a priori tolerance for $N = 10$ and $\beta = 0.1$. Evaluating the step function at the declared value $\tau = 0.01$ gives $M_\star = 3$.

The plateau structure also makes the sensitivity of the decision visible. The selected dimension changes only when the a priori tolerance crosses a canonical tail weight; small tolerance perturbations within the same plateau leave the reduction unchanged.

4.2 Algebraic closure and continuum accuracy separate cleanly

Figure 3 reports every scalar diagnostic defined in Section 3.2. The six curves remain in the binary64 roundoff regime throughout the grid sweep, but they do not behave identically. Spectral reconstruction is the largest of the full-space checks because it accumulates perturbations over an expanding matrix. Within the retained sector, leakage grows most visibly with the finite-difference energy scale, whereas the projector-idempotency defect remains essentially flat. This pattern is the expected signature of unnormalized algebraic residuals: their absolute magnitudes inherit representation scale even when the underlying identities continue to close numerically.

The normalized eigensolver residuals in Figure 4 remove that representation scale. All three norm ratios remain at roundoff, with no convergence curve to interpret. Their small norm-dependent fluctuations show solver arithmetic rather than a physical or continuum trend. Taken together,

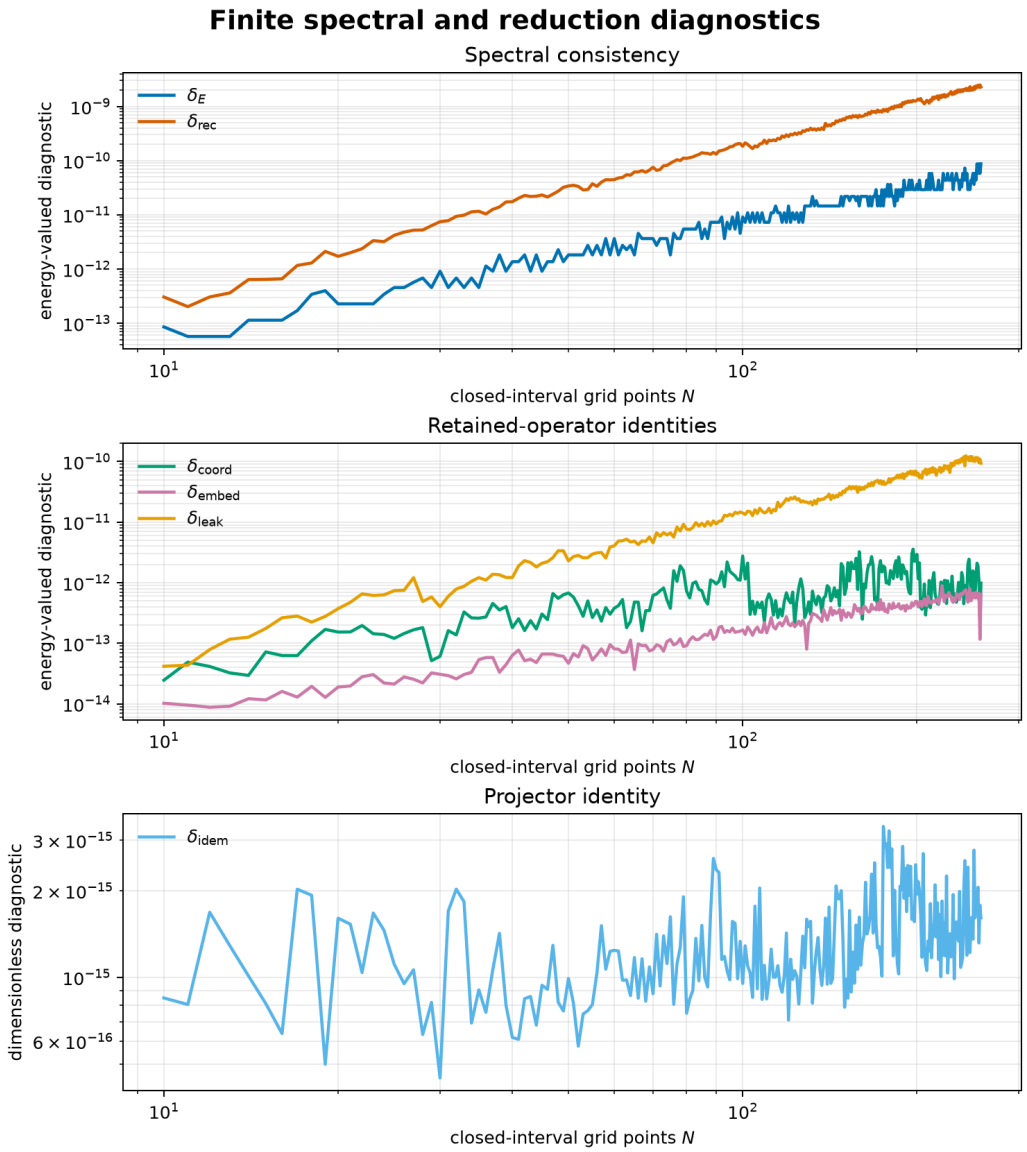


Figure 3: The six finite spectral and reduction diagnostics for every closed-interval grid size $N = 10, 11, \ldots, 258$, with retained dimension $M = 3$. The curves correspond directly to Equations (37)–(42). They test represented finite-dimensional identities and do not measure continuum discretization error.

Figures 3 and 4 establish that the numerical eigensystem is internally consistent before any comparison with the continuum model is made.

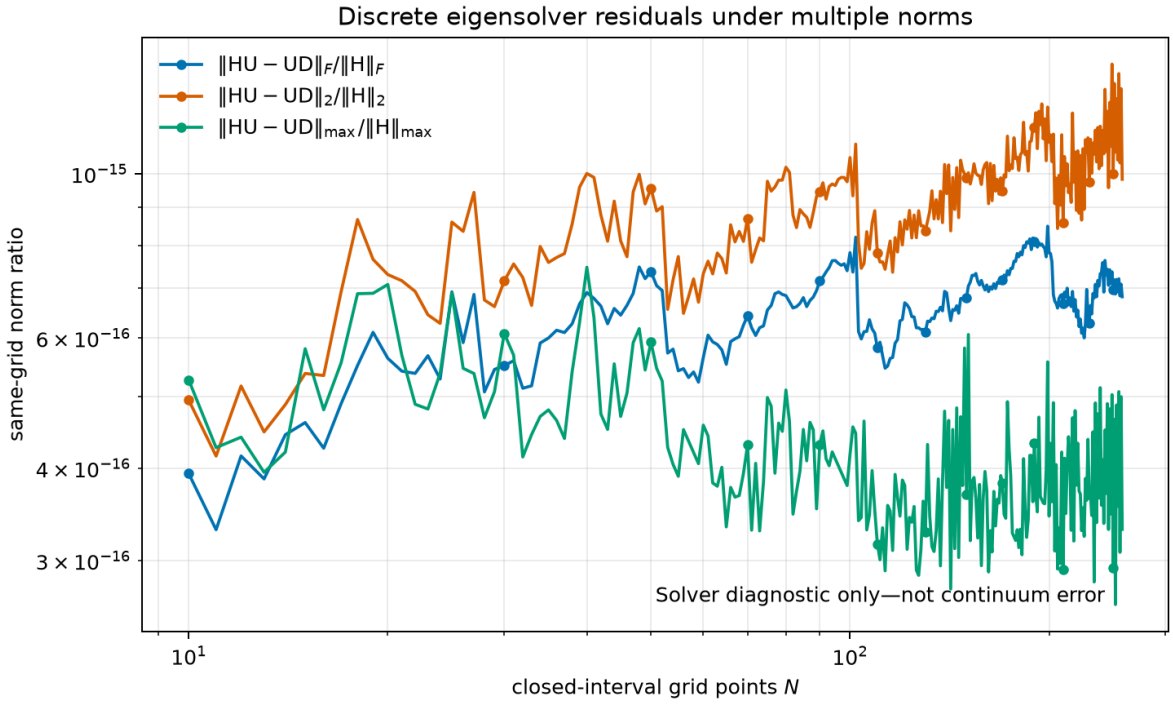


Figure 4: Full-eigensystem algebraic residuals under the Frobenius, spectral, and maximum-entry norms. Matrices are shown in upright sans-serif notation in the figure. The plotted quantities test $\mathsf{H}_{\Delta x}\mathsf{U} - \mathsf{U}\widetilde{\mathsf{D}}_{\Delta x}$ and do not measure continuum error.

Figure 5 then exposes a different phenomenon. The retained vectors have the expected nodal sine structure, while the finite-difference energy ladder bends progressively below the quadratic continuum ladder. The eigenvectors can therefore be correct at the grid nodes even when their associated continuum energies are biased. That separation between nodal geometry and energetic dispersion is the central feature of this calibration problem.

The mode-resolution map in Figure 6 makes the same structure systematic. Every grid produces a rising error profile across mode index, and refinement shifts the entire profile downward. Resolution is therefore controlled by both N and the wavelength being represented; a grid that is accurate for the ground state need not be accurate for the upper part of even a small retained window.

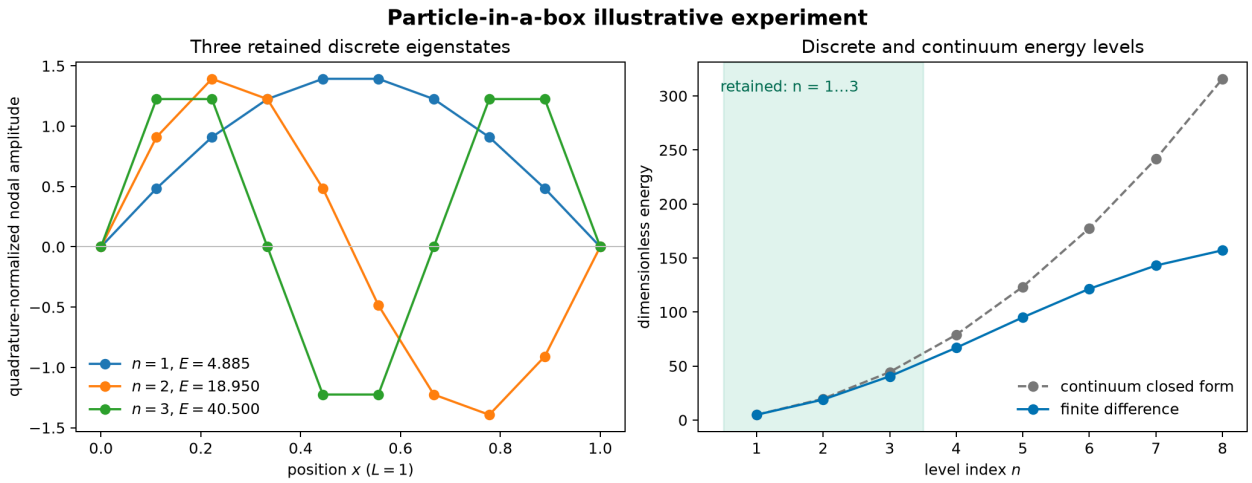


Figure 5: The three retained discrete eigenstates and the finite and continuum energy levels for $N = 10$ (eight interior points). Accurate solution of the discrete matrix does not remove its finite-difference dispersion.

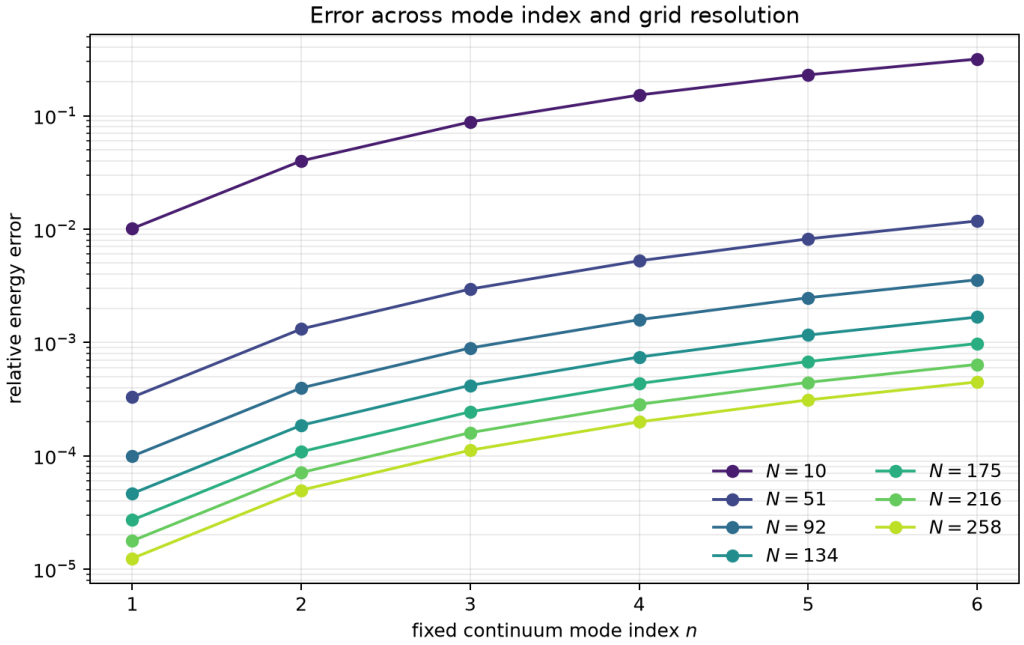


Figure 6: Relative continuum-energy error across modes $n = 1, \ldots, 6$ for representative closed-interval grids. The underlying calculation includes every integer grid size from $N = 10$ through $N = 258$.

4.3 Convergence is modewise, not uniform over the spectrum

The fixed low modes in Figure 7 follow parallel second-order trajectories. Their observed orders approach two from below, while the mode-dependent vertical offsets record the increasing dispersion penalty for shorter wavelengths. Refinement therefore behaves exactly as the centered stencil predicts when the mode index is held fixed.

The conclusion changes when the mode index grows with the grid. In Figure 8, spectra from widely different matrix dimensions collapse onto one dispersion curve when expressed in terms of

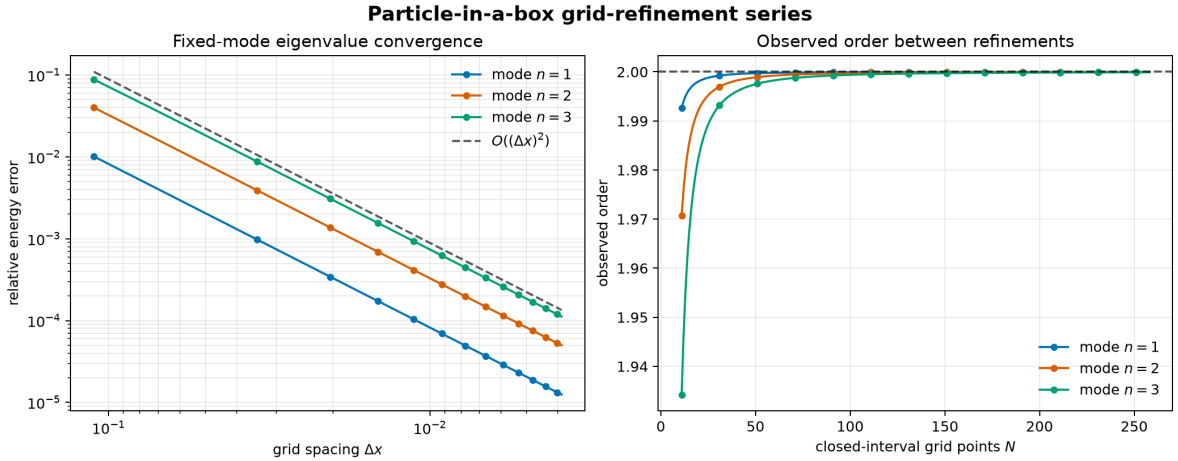


Figure 7: Relative energy error and observed refinement order for the first three continuum modes. Each fixed mode approaches second-order convergence.

the fractional index $q = n/(N-1)$. Refinement adds higher-frequency modes but does not reduce the error at a fixed nonzero q. Near the top of the spectrum the discrete energy remains far below the continuum value, even though the nodal-overlap defects stay at roundoff. The plot therefore separates two statements that a single residual cannot: the matrix eigenproblem is solved accurately, but the stencil does not approximate all resolved wavelengths uniformly.

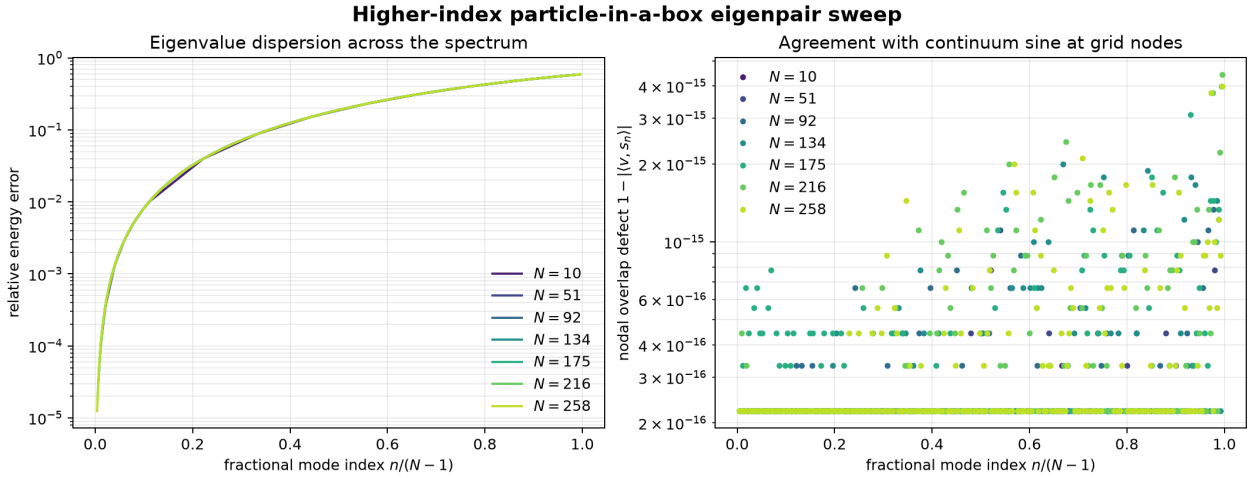


Figure 8: Full-spectrum diagnostics for representative members of the dense grid sweep. The left panel shows finite-difference energy dispersion versus fractional mode index. The right panel shows nodal wavefunction-overlap defects.

The higher fixed modes in Figure 9 reconcile these two views. Each eventually approaches second order, but the onset is delayed as the mode index increases. Second-order convergence is thus an asymptotic statement about a fixed physical mode, not a blanket statement about every eigenvalue returned by a growing discretization.

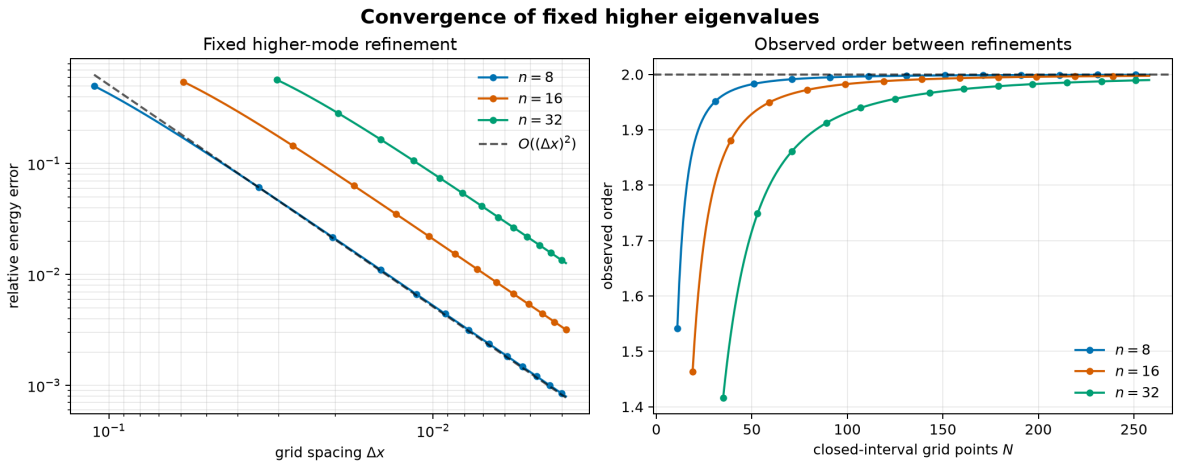


Figure 9: Relative energy error and observed refinement order for selected fixed higher modes. The second-order regime appears later as the mode index increases.

4.4 Residual geometry records the comparison being made

For the box model, the finite Hamiltonian is the finite Dirichlet kinetic matrix, $\mathsf{H}_{\Delta x} = \mathsf{T}_{\mathrm{D},\Delta x}$. The consistently compressed residual defined in Equation (45) therefore satisfies

$$\mathsf{R}_{\mathrm{cons}} = 0. \tag{52}$$

This is the reduced physical potential for the declared finite model. The blank left panel of Figure 10 is therefore a substantive result: applying the same projection to both terms preserves the exact zero-potential identity.

The unmatched residual in Equation (46) instead satisfies

$$\mathsf{R}_{\mathrm{un}} = -\mathsf{Q}_M \mathsf{T}_{\mathrm{D},\Delta x} \mathsf{Q}_M. \tag{53}$$

The middle panel of Figure 10 consequently reproduces the structured discarded kinetic sector. Its nonzero diagonal and near-diagonal pattern is not a newly inferred potential; it is the visible consequence of subtracting an uncompressed operator from a compressed one.

The right panel isolates a third geometry. Changing from the Dirichlet to the cyclic closure modifies only the two corner couplings, so the residual is sparse and boundary-localized. The residual therefore isolates the effect of changing the boundary condition in the discrete operator; it is not a potential-energy term.

Figure 11 shows why the heatmap geometry must accompany a scalar norm. With fixed M, the unmatched residual contains the entire high-energy sector. Its normalized spectral norm is therefore one, and its Frobenius and maximum-entry ratios approach one as the grid grows: almost all of the expanding operator lies outside the fixed retained subspace. The boundary residual behaves in the opposite way. Its two corner entries become negligible in the global Frobenius ratio, while its spectral ratio approaches one quarter and its maximum-entry ratio remains one half. No contradiction is present; the three norms ask different questions of the same sparse boundary perturbation.

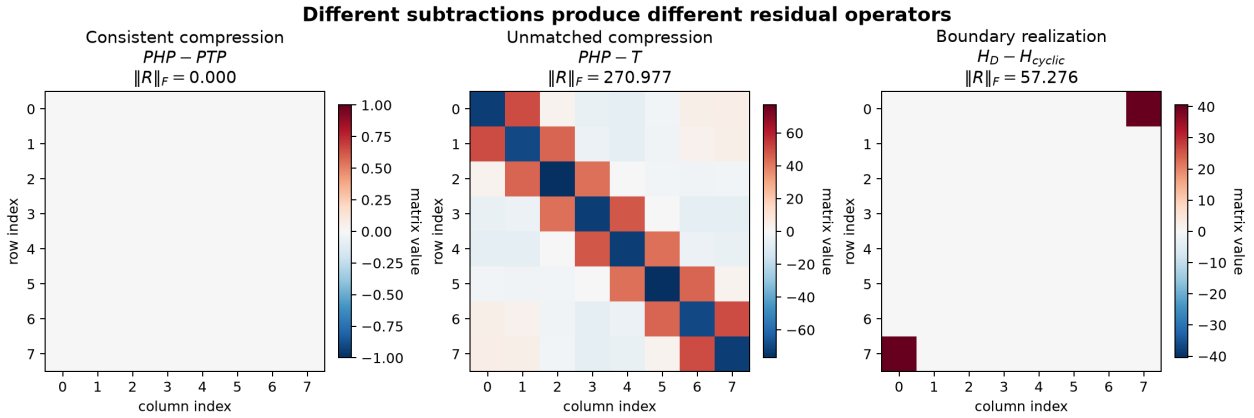


Figure 10: Three residual constructions. Consistent compression gives zero; unmatched compression retains the discarded kinetic sector; changing the boundary closure changes two corner couplings.

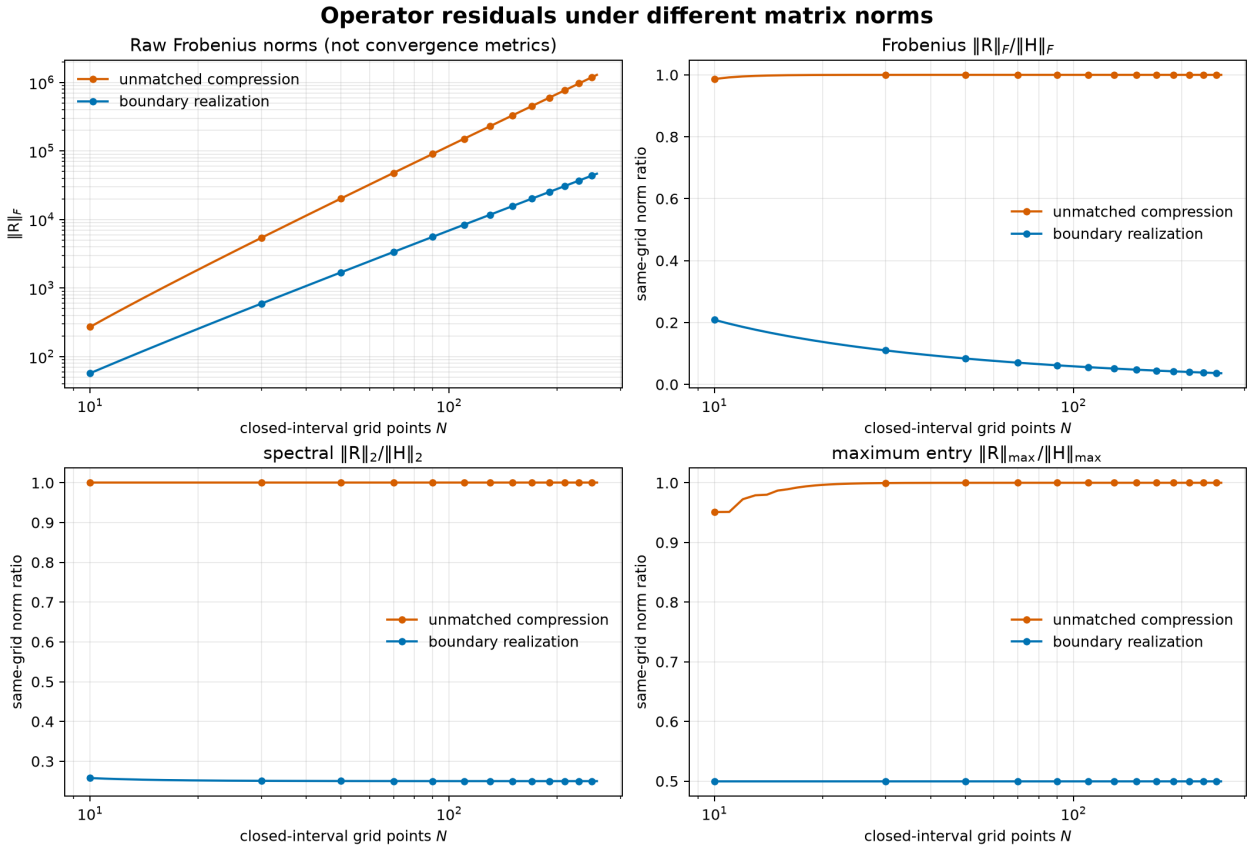


Figure 11: Raw and same-grid-normalized residual norms for every closed-interval grid size $N = 10, 11, \ldots, 258$. The panels demonstrate that norm choice is part of the diagnostic definition.

4.5 Identifiability comes from the model class

Let $\mathcal{K}$ and $\mathcal{V}$ be admissible classes for the kinetic and potential parts of a matrix H. The compatible decompositions form the fiber

$$\mathcal{D}_{\mathsf{H}}(\mathcal{K}, \mathcal{V}) = \{(\mathsf{T}, \mathsf{V}) : \mathsf{T} \in \mathcal{K},\ \mathsf{V} \in \mathcal{V},\ \mathsf{H} = \mathsf{T} + \mathsf{V}\}. \tag{54}$$

A sufficient global uniqueness condition is

$$(\mathcal{K} - \mathcal{K}) \cap (\mathcal{V} - \mathcal{V}) = \{0\}. \tag{55}$$

Without such a restriction, the matrix alone does not select one additive split.

The retained calculation starts from $\mathsf{H}_{\mathrm{red}} = \mathsf{T}_{\mathrm{D,red}} + 0$. For a declared real-symmetric shift K, the same matrix also satisfies

$$\mathsf{H}_{\mathrm{red}} = (\mathsf{T}_{\mathrm{D,red}} - \mathsf{K}) + \mathsf{K}. \tag{56}$$

The equality is algebraically exact, but it does not make K a physical potential. Figure 12 shows what the successive model classes contribute. Scalar and diagonal classes leave most of the illustrative shift unexplained because they cannot represent its off-diagonal structure. Allowing nearest-neighbor couplings captures most of that structure, while the unrestricted real-symmetric class fits exactly by construction. The monotone reduction in residual is therefore not evidence that the unrestricted decomposition is more physical. It records only the increasing expressiveness of the admissible class; physical identifiability requires that class to be justified independently.

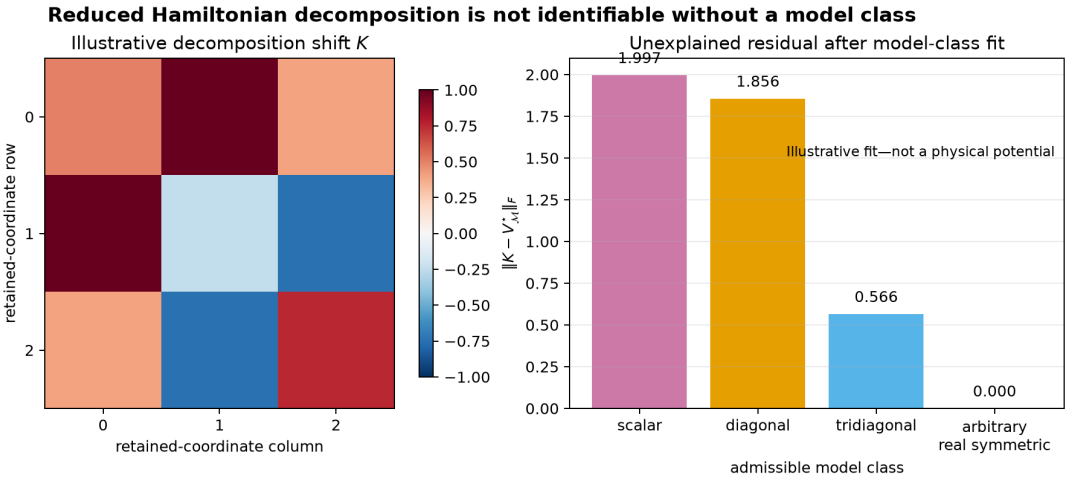


Figure 12: An illustrative decomposition shift and its best fit in four model classes. Exact representability in the unrestricted class does not establish a physical interpretation.

4.6 Implications for reduced-model diagnostics

Taken together, the results establish three distinct levels of assessment. Algebraic closure asks whether the represented eigenproblem, projector, and compression identities are satisfied. Continuum approximation asks whether the represented modes resolve the wavelengths of interest. Physical interpretation asks whether a residual or additive decomposition belongs to a justified model class. Success at one level does not establish success at the next.

The residual study also shows that a norm cannot supply meaning that is absent from the operator being measured. Consistent compression, unmatched compression, and a change of boundary condition produce different residual matrices before any norm is chosen. Frobenius, spectral, and maximum-entry norms then expose different global, action-based, and coordinate-local properties of those matrices. Both the residual definition and its representation must therefore precede scalar comparison.

PIAB1D is a calibration case because its retained trial space is an exact spectral subspace. More general trial spaces need not be invariant, so projector defects, leakage, analytical residuals, and

operator-comparison errors can become independent channels rather than roundoff-level identities. The separation demonstrated here provides a controlled reference for interpreting those channels in less structured models.

5 Verification, formalization, and limitations

The calculation is checked at three distinct levels. JSON schemas check document structure. Numerical verification scripts recompute finite formulas and algebraic relationships from retained data. Formal numerical analysis can extend much farther: prior work has mechanized a centered finite-difference wave solver from analytical error through its C implementation [3, 4], and the Lax equivalence theorem for finite-difference schemes has been formalized in Coq [25]. The separate Lean 4 development used here has a more limited target. It uses mathlib [5, 27] to prove exact statements about finite Dirichlet recurrences, matrix eigenpairs, orthogonal compression, decomposition nonuniqueness, and finite model-class fitting.

These levels are complementary but not interchangeable. The Lean proofs do not certify NumPy or SciPy execution, serialized bytes, binary64 rounding, plotted figures, or the continuum limit. The numerical calculations do not prove general theorems. The verification scripts are separate code paths in the same repository; they are not external replications.

The model itself is also intentionally limited. No experimental system is identified, no parameters are calibrated to a material, and no uncertainty model is introduced. Agreement with the analytical particle-in-a-box spectrum verifies the implementation against the same mathematical model. It is not physical validation.

6 Software and data availability

The software and documentation supporting this work are maintained in `ksdft2effmass`. The calculation uses components from the ProjectKoios system of libraries developed by the author, including `projectkoios-physkit` and `projectkoios-optimization`, whose `projectkoios.numerics.quadrature` package supplies the unit-free rule. These repositories are publicly available under the Apache License 2.0.

The demonstrations, retained numerical results, verification programs, and plotting sources are preserved together under the stable calculation owner

`https://github.com/eragasa/ksdft2effmass/tree/main/calculations/piab1d`

File-level provenance and byte integrity are recorded in the accompanying `SHA256SUMS` manifest. At release, the archived repository snapshot will also be identified by its immutable Git commit identifier. The Git commit identifies the source snapshot, whereas the SHA-256 manifest identifies the retained artifact bytes.

7 Conclusion

The one-dimensional particle in a box gives an unusually clean view of spectral reduction. The finite matrix is solved accurately, its lowest eigenspace is invariant, and its reduced Hamiltonian is known exactly in spectral coordinates. The remaining spectral discrepancy is finite-difference dispersion: it is second order for fixed resolved modes but remains large near the top of a growing spectrum.

The same calculation also shows why residual operators require careful definitions. Consistent compression, unmatched compression, and a change of boundary closure lead to different matrices

with different interpretations. An additive decomposition is similarly meaningful only relative to an admissible model class. The contribution is the integration of these distinctions with independently recomputed numerical artifacts and a bounded formal crosswalk, not novelty of the box solution, the stencil, or spectral projection. As the first stage of the planned diagnostic sequence, this calculation establishes the reference case against which later spatial truncation, periodic, lattice, multiband, and first-principles reductions can be interpreted.

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Supplemental material

A Exact finite identities underlying the diagnostics

This appendix proves the finite identities used to interpret the numerical figures. The proofs concern exact real matrices and exact scalar formulas. They do not certify the binary64 calculations, serialized results, or plotting programs.

A.1 Monotonicity of the retained canonical weight

Let $d = N - 2$ and define the represented canonical probabilities

$$p_n^{(\beta,\Delta x)} = \frac{\exp(-\beta E_{n,\Delta x})}{Z_{\Delta x}}, \qquad n = 1, \dots, d. \tag{57}$$

Every probability is strictly positive, and $w_{>M}^{(\beta,\Delta x)} = \sum_{n=M+1}^{d} p_n^{(\beta,\Delta x)}$. For $M < d$,

$$w_{>(M+1)}^{(\beta,\Delta x)} = w_{>M}^{(\beta,\Delta x)} - p_{M+1}^{(\beta,\Delta x)} < w_{>M}^{(\beta,\Delta x)}. \tag{58}$$

Moreover, $w_{>d}^{(\beta,\Delta x)} = 0$. Hence the feasible set in Equation (51) is nonempty for every $\tau \geq 0$, and its smallest element is unique. If $\tau_1 \leq \tau_2$, every dimension feasible for τ_1 is feasible for τ_2, so

$$M_\star(N, \beta, \tau_2) \leq M_\star(N, \beta, \tau_1). \tag{59}$$

Thus $M_\star$ is a nonincreasing integer-valued step function of the a priori tolerance. Its transitions occur at the finite set of canonical tail weights. □

Scientific interpretation. The inverse temperature β identifies the energy range populated by the declared canonical ensemble, while τ specifies how much of that population may be omitted. The rule therefore converts a scientific resolution requirement into the smallest state count that satisfies it. A plateau means that the selected model size is stable to changes of τ within that interval; it does not make the choice independent of the ensemble or tolerance.

A.2 Exact eigensystem of the Dirichlet difference matrix

Set $d = N - 2$, $\Delta x = L/(d+1)$, and

$$a = \frac{\hbar^2}{2m(\Delta x)^2}. \tag{60}$$

Let T_d be the $d \times d$ matrix with diagonal entries 2, first off-diagonal entries -1, and all other entries zero, so that $\mathsf{H}_{\Delta x} = a\mathsf{T}_d$. For $n = 1, \dots, d$, define

$$\theta_n = \frac{n\pi}{d+1}, \qquad (\boldsymbol{u}_n)_j = \sqrt{\frac{2}{d+1}} \sin(j\theta_n), \quad j = 1, \dots, d. \tag{61}$$

The ghost values at $j = 0$ and $j = d + 1$ vanish. At every interior coordinate,

$$(\mathsf{T}_d \boldsymbol{u}_n)_j = 2(\boldsymbol{u}_n)_j - (\boldsymbol{u}_n)_{j-1} - (\boldsymbol{u}_n)_{j+1} \tag{62}$$

$$= 2(1 - \cos\theta_n)(\boldsymbol{u}_n)_j \tag{63}$$

$$= 4\sin^2\left(\frac{\theta_n}{2}\right)(\boldsymbol{u}_n)_j. \tag{64}$$

The discrete sine orthogonality identity

$$\sum_{j=1}^{d} \sin(j\theta_n)\sin(j\theta_m) = \frac{d+1}{2}\,\delta_{nm} \tag{65}$$

follows by applying the product-to-sum formula and evaluating the two finite cosine sums. Therefore the vectors in Equation (61) form an orthonormal basis, and

$$\mathsf{H}_{\Delta x}\boldsymbol{u}_n = E_{n,\Delta x}\boldsymbol{u}_n, \qquad E_{n,\Delta x} = 4a\sin^2\!\left(\frac{n\pi}{2(d+1)}\right). \tag{66}$$

Writing $\mathsf{U} = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_d]$ and $\mathsf{D}_{\Delta x} = \mathrm{diag}(E_{1,\Delta x}, \ldots, E_{d,\Delta x})$ gives the exact reconstruction

$$\mathsf{U}^{\mathsf{T}}\mathsf{U} = \mathsf{I}, \qquad \mathsf{H}_{\Delta x} = \mathsf{U}\mathsf{D}_{\Delta x}\mathsf{U}^{\mathsf{T}}. \tag{67}$$

□

Scientific interpretation. The vectors $\boldsymbol{u}_n$ are the standing-wave shapes permitted by two impenetrable ends, sampled on the interior grid. Their increasing node count corresponds to shorter wavelength and higher kinetic energy. Orthogonality makes these modes independent coordinates of the finite model, and reconstruction shows that they exhaust that model. The result is exact for the discrete Hamiltonian; it does not by itself establish that every discrete energy accurately represents the continuum box.

A.3 Exact identities for spectral reduction

Let $\mathsf{U}_M = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_M]$, $\mathsf{D}_M = \mathrm{diag}(E_{1,\Delta x}, \ldots, E_{M,\Delta x})$, and $\mathsf{P}_M = \mathsf{U}_M\mathsf{U}_M^{\mathsf{T}}$. Orthonormality and $\mathsf{H}_{\Delta x}\mathsf{U}_M = \mathsf{U}_M\mathsf{D}_M$ imply

$$\mathsf{P}_M^2 = \mathsf{U}_M(\mathsf{U}_M^{\mathsf{T}}\mathsf{U}_M)\mathsf{U}_M^{\mathsf{T}} = \mathsf{P}_M, \tag{68}$$

$$\mathsf{H}_{\mathrm{red}} = \mathsf{U}_M^{\mathsf{T}}\mathsf{H}_{\Delta x}\mathsf{U}_M = \mathsf{D}_M, \tag{69}$$

$$\mathsf{P}_M\mathsf{H}_{\Delta x}\mathsf{P}_M = \mathsf{U}_M\mathsf{D}_M\mathsf{U}_M^{\mathsf{T}}. \tag{70}$$

For $\mathsf{Q}_M = \mathsf{I} - \mathsf{P}_M$,

$$\mathsf{Q}_M\mathsf{H}_{\Delta x}\mathsf{U}_M = (\mathsf{I} - \mathsf{P}_M)\mathsf{U}_M\mathsf{D}_M = \mathsf{0}. \tag{71}$$

Equations (67)– (71) show that δ_{rec}, δ_{idem}, δ_{coord}, δ_{embed}, and δ_{leak} vanish in exact arithmetic. Their nonzero calculated values therefore measure finite-precision closure of these identities, not continuum error. □

Scientific interpretation. Retaining exact eigenvectors selects an invariant low-energy sector of the represented box. Applying the finite Hamiltonian to a retained state cannot generate a component outside that sector, so the leakage is exactly zero before floating-point evaluation. This makes PIAB1D a calibration case: nonzero leakage in a general trial basis would measure genuine coupling to omitted represented states, whereas here it can only measure numerical closure. Neither outcome determines the continuum discretization error.

A.4 Pointwise convergence and full-spectrum dispersion

The continuum energy can be written using the same scale a as

$$E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2} = 4az_{n,d}^2, \qquad z_{n,d} = \frac{n\pi}{2(d+1)}. \tag{72}$$

Combining this expression with Equation (66) gives

$$\frac{E_{n,\Delta x}}{E_n} = \left(\frac{\sin z_{n,d}}{z_{n,d}}\right)^2, \qquad \epsilon_{n,N} = 1 - \left(\frac{\sin z_{n,d}}{z_{n,d}}\right)^2. \tag{73}$$

For fixed n, the Taylor expansion at the origin yields

$$\epsilon_{n,N} = \frac{z_{n,d}^2}{3} + O(z_{n,d}^4) = \frac{n^2\pi^2}{12L^2}(\Delta x)^2 + O((\Delta x)^4). \tag{74}$$

This proves second-order convergence for every fixed mode. In contrast, suppose that $n/(d+1) \to q$ with $q \in (0, 1]$. Then

$$\epsilon_{n,N} \longrightarrow 1 - \left[\frac{\sin(\pi q/2)}{\pi q/2}\right]^2 > 0, \tag{75}$$

because $\sin x < x$ for $x > 0$. Thus refinement does not give uniform relative spectral convergence over a mode set whose dimension grows with the grid. This is the exact curve onto which the full-spectrum results collapse. □

Scientific interpretation. Holding n fixed means holding a physical standing-wave pattern fixed while placing more grid points across it; the wavelength is increasingly well resolved. Holding $q = n/(d+1)$ fixed instead keeps the wavelength at a fixed fraction of the grid scale, so the number of points per oscillation does not improve. Grid refinement then adds new short-wavelength states without making those states better continuum approximations. This is why low-energy observables may converge even though the full growing spectrum does not converge uniformly.

A.5 Compression and boundary residuals

Because $\mathsf{H}_{\Delta x} = \mathsf{T}_{\mathrm{D},\Delta x}$, consistent compression vanishes immediately. Spectral projectors commute with the Hamiltonian, so

$$\mathsf{H}_{\Delta x} = \mathsf{P}_M \mathsf{H}_{\Delta x} \mathsf{P}_M + \mathsf{Q}_M \mathsf{H}_{\Delta x} \mathsf{Q}_M. \tag{76}$$

The unmatched residual is therefore

$$\mathsf{R}_{\mathrm{un}} = -\mathsf{Q}_M \mathsf{H}_{\Delta x} \mathsf{Q}_M = -\sum_{n=M+1}^{d} E_{n,\Delta x} \boldsymbol{u}_n \boldsymbol{u}_n^{\mathsf{T}}. \tag{77}$$

Since the energies are positive and increasing,

$$\|\mathsf{R}_{\mathrm{un}}\|_2 = E_{d,\Delta x}, \qquad \|\mathsf{R}_{\mathrm{un}}\|_{\mathrm{F}}^2 = \sum_{n=M+1}^{d} E_{n,\Delta x}^2. \tag{78}$$

Hence the Hamiltonian-normalized spectral norm is exactly one whenever $M < d$. For fixed M, the normalized Frobenius norm approaches one as d grows because the finite retained sum becomes negligible relative to the full spectral square sum.

For the cyclic reference, let $\boldsymbol{e}_1$ and $\boldsymbol{e}_d$ denote the first and last coordinate vectors and assume $d \geq 2$. The cyclic matrix adds the two corner couplings $-a$, so

$$\mathsf{H}_{\mathrm{cyclic},\Delta x} = \mathsf{H}_{\mathrm{D},\Delta x} - a(\boldsymbol{e}_1\boldsymbol{e}_d^{\mathsf{T}} + \boldsymbol{e}_d\boldsymbol{e}_1^{\mathsf{T}}), \tag{79}$$

and therefore

$$\mathsf{R}_{\mathrm{bc}} = a(\boldsymbol{e}_1\boldsymbol{e}_d^{\mathsf{T}} + \boldsymbol{e}_d\boldsymbol{e}_1^{\mathsf{T}}). \tag{80}$$

This matrix has two nonzero entries and acts on $\mathrm{span}\{\boldsymbol{e}_1, \boldsymbol{e}_d\}$ as $\left[\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right]$. Consequently,

$$\|\mathsf{R}_{\mathrm{bc}}\|_{\mathrm{F}} = \sqrt{2}\,a, \qquad \|\mathsf{R}_{\mathrm{bc}}\|_2 = a, \qquad \|\mathsf{R}_{\mathrm{bc}}\|_{\max} = a. \tag{81}$$

Counting the entries of the Dirichlet tridiagonal matrix and using its largest eigenvalue gives

$$\|\mathsf{H}_{\Delta x}\|_{\mathrm{F}} = a\sqrt{6d-2}, \quad \|\mathsf{H}_{\Delta x}\|_2 = 4a\cos^2\!\left(\frac{\pi}{2(d+1)}\right), \quad \|\mathsf{H}_{\Delta x}\|_{\max} = 2a. \tag{82}$$

The normalized boundary diagnostics are therefore

$$\eta_{\mathrm{F}}(\mathsf{R}_{\mathrm{bc}}) = \frac{1}{\sqrt{3d-1}}, \qquad \eta_2(\mathsf{R}_{\mathrm{bc}}) = \frac{1}{4\cos^2(\pi/[2(d+1)])}, \qquad \eta_{\max}(\mathsf{R}_{\mathrm{bc}}) = \frac{1}{2}. \tag{83}$$

Thus the Frobenius ratio tends to zero, the spectral ratio tends to one quarter, and the maximum-entry ratio remains one half. The distinct trends in Figure 11 follow from the matrix geometry rather than from conflicting numerical results. □

Scientific interpretation. The unmatched residual is dominated by the high-energy states deliberately excluded from the reduced model; calling it a potential would mistake omitted kinetic content for new physics. The boundary residual has a different origin. Connecting the first and last coordinates changes an open box into a finite ring-like closure, and the two corner entries record only that boundary choice. Its norm describes the size of the boundary modification in a chosen matrix metric, not the strength of a continuum potential.

A.6 Identifiability and Frobenius-optimal model classes

Suppose $\mathsf{H} = \mathsf{T}_1 + \mathsf{V}_1 = \mathsf{T}_2 + \mathsf{V}_2$ with $\mathsf{T}_1, \mathsf{T}_2 \in \mathcal{K}$ and $\mathsf{V}_1, \mathsf{V}_2 \in \mathcal{V}$. Rearrangement gives

$$\mathsf{T}_1 - \mathsf{T}_2 = \mathsf{V}_2 - \mathsf{V}_1 \in (\mathcal{K} - \mathcal{K}) \cap (\mathcal{V} - \mathcal{V}). \tag{84}$$

If the intersection is $\{\mathsf{0}\}$, then $\mathsf{T}_1 = \mathsf{T}_2$ and $\mathsf{V}_1 = \mathsf{V}_2$. This proves the uniqueness condition in Equation (55). Conversely, any nonzero matrix in the intersection supplies two distinct admissible decompositions of the same sum, so the condition also characterizes global uniqueness over the declared classes.

The model-class fits in Figure 12 are orthogonal projections in the Frobenius inner product. For a coordinate-support class $\mathcal{C}$, let $\Pi_{\mathcal{C}}\mathsf{K}$ retain the permitted entries and set all others to zero. The supported and omitted entries are disjoint, so every $\mathsf{V} \in \mathcal{C}$ satisfies the Pythagorean identity

$$\|\mathsf{K} - \mathsf{V}\|_{\mathrm{F}}^2 = \|\mathsf{K} - \Pi_{\mathcal{C}}\mathsf{K}\|_{\mathrm{F}}^2 + \|\Pi_{\mathcal{C}}\mathsf{K} - \mathsf{V}\|_{\mathrm{F}}^2. \tag{85}$$

Hence the diagonal and tridiagonal minimizers are obtained by retaining, respectively, the diagonal entries and the entries with $|i-j| \leq 1$. For the scalar-identity class,

$$\|\mathsf{K} - c\mathsf{I}\|_{\mathrm{F}}^2 = \sum_i (K_{ii} - c)^2 + \sum_{i \neq j} K_{ij}^2 \tag{86}$$

is minimized at $c = \mathrm{tr}(\mathsf{K})/M$. Finally, because the illustrative shift is itself real symmetric, the unrestricted real-symmetric class contains K and has zero residual. These are statements about approximation within the declared matrix classes; they do not assign physical meaning to the fitted matrices. □

Scientific interpretation. An observed reduced Hamiltonian does not reveal, by itself, which entries should be called kinetic and which should be called potential. A more expressive matrix class will always fit at least as well, and an unrestricted class fits exactly by definition. Scientific identification therefore requires independently justified structure—for example locality, symmetry, known interactions, or a prescribed basis—rather than fit quality alone.

B Machine-checked crosswalk

The companion formalization is maintained in `github.com/eragasa/projectkoios-formal`. The calculation sources are maintained in `github.com/eragasa/ksdft2effmass`. Table 2 identifies the machine-checked theorem families corresponding to the exact finite arguments above. The GitHub formalization and this numerical repository remain separate evidence owners.

Appendix statement	Formal theorem identifiers	Boundary of the formal result
Canonical finite Dirichlet mode and discrete eigenvalue	PIAB1D-SPECTRUM-THM-015, PIAB1D-SPECTRUM-THM-016	Exact finite eigenpair; no binary64 or continuum-completeness claim
Dirichlet–cyclic corner relation and support	PIAB1D-BOUNDARY-THM-001, PIAB1D-BOUNDARY-THM-002, PIAB1D-BOUNDARY-THM-004	Exact finite matrix identity and periodic specialization
Consistent and unmatched orthogonal compression	QM-COMPRESSION-THM-001, QM-COMPRESSION-THM-002	Exact under the declared idempotency and commutation hypotheses
Shifted decomposition nonuniqueness	QM-IDENTIFIABILITY-THM-001, QM-IDENTIFIABILITY-THM-002	Exact preservation of the sum and distinctness for a nonzero shift
Frobenius-optimal supported model classes	QM-MODEL-FIT-THM-001–QM-MODEL-FIT-THM-006	Exact finite sums and coordinate-support fits
Fixed-mode Taylor order, fractional-mode limit, and canonical selection rule	Manuscript derivations in this appendix	Not currently claimed as machine-checked declarations

Table 2: Crosswalk between the human-readable appendix and the companion Lean formalization. The theorem identifiers name exact finite declarations, not runtime or scientific-validation certificates.